

Option C – The Physics of Sports

Examiner
only

8. (a) Define angular acceleration. [2]

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(b) A gymnast begins a routine to dismount from the horizontal bar by increasing her angular velocity from 3.4 rad s^{-1} to 8.0 rad s^{-1} in a time of 2.3 s . The moment of inertia of the gymnast is 34 kg m^2 .



(i) Calculate the torque on the gymnast. [4]

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- (ii) The gymnast lets go of the bar and somersaults in the air before she lands on her feet on the ground. Explain why the gymnast pulls in her arms and adopts a tuck position as she is in the air. [4]

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- (iii) The gymnast, of mass 52 kg, lands on the mat with a velocity of 5.7 m s^{-1} and comes to rest in a time of 0.26 s. Show clearly that the mean force exerted by the gymnast on the mat is approximately 1000 N. [3]

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- (c) In a separate event, a male gymnast performs a routine on the rings as shown.



- (i) Explain in terms of centre of gravity why he does not rotate. [2]

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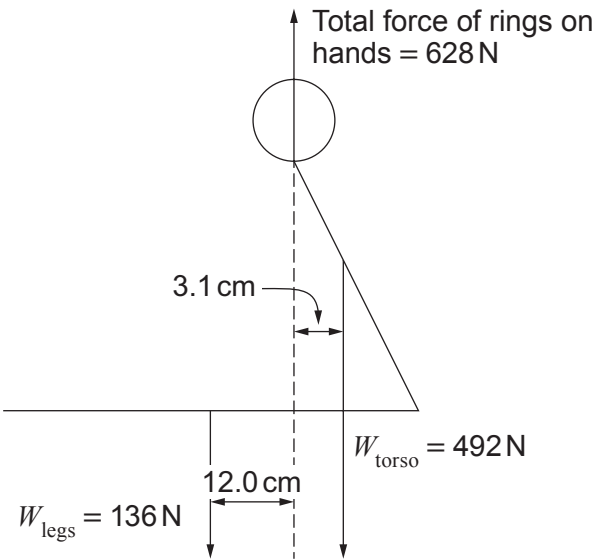
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- (ii) At a later point the forces acting on the gymnast are shown in the following diagram. Justify, using calculations, the motion of the gymnast. [5]



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